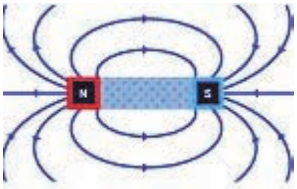
CHAPTER 12

MAGNETIC EFFECTS OF ELECTRIC CURRENT



You have studied about electricity and magnetism in your previous classes. You would be surprised to know that both of them are effects produced by a single phenomenon, i.e. wherever an electric charge is active, it produces a magnetic effect. Also, wherever a magnetic field exists, electric charge in some form of other will be preswent. You are familiar with magnets and the magnetic compass.

You have studied about the various properties of a magnet in previous classes. List the various properties that a magnet exhibits.

12.1 Magnetic field and magnetic lines of force

Place a small magnetic compass near a bar magnet. The compass needle is deflected and becomes stationary pointing in a particular direction. This shows that a force has acted upon the compass needle, due to which it moved in a particular direction before coming to a stop.

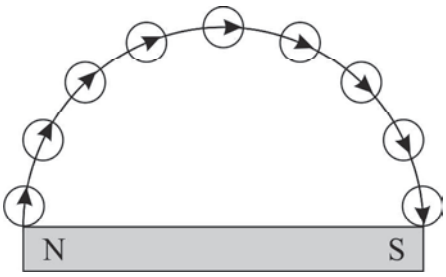


Figure-1: magnetic lines of force

The area surrounding a magnet, in which a force is exerted on the magnetic compass due to which it turns and stops in a particular direction, is known as the magnetic field. The magnetic field of a magnet is shown by imaginary magnetic lines of force surrounding it.

In a magnetic field, the direction of magnetic compass needle changes its constantly on moving it from one point to another. When the compass is moved from the magnet's North Pole towards its South Pole, it forms a curved path, which is known as magnetic lines of force.

At any point on this curved line of force, if a line tangential to that point is drawn, it will indicate the direction of the magnetic line of force at that point.

Magnetic lines of force have the following properties:

1. They originate at the North Pole and travel to the South Pole in the area surrounding the magnet.
2. They continue from the South Pole to the North Pole inside the magnet.
3. They form a closed loop.
4. The lines of force do not cross each other at any point.
5. Magnetic lines of force are concentrated at the poles and spaced apart at other sections of the field, which indicates that field intensity is strongest at the poles.

You can observe the magnetic field by conducting the following activity.

Activity-1

Items required: Drawing board, white paper, drawing pins, bar magnet, iron filings (or powder).

Method:

1. Fix a white paper on the drawing boards using the pin.
2. Place a bar magnet at the centre of this paper.
3. Now sprinkle the iron-filings evenly around the magnet.
4. Now gently tap the drawing board with your fingers. Note your observations.

You will observe that the magnet exerts a force on the iron filings which then arrange themselves in a particular shape, as shown in figure-2.

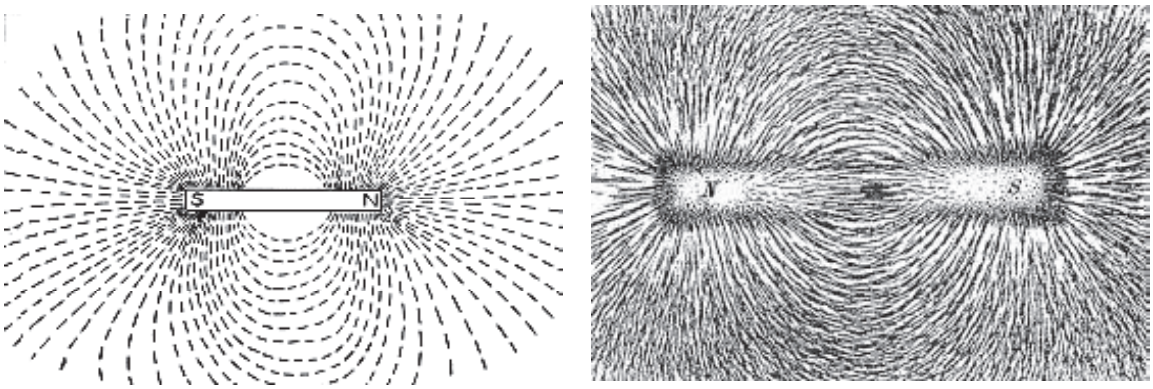


Figure-2: Magnetic fields

The area surrounding a magnet where the magnetic force is observable is known as the magnetic field of the magnet. The iron filings arrange themselves in a particular pattern, and these are known as magnetic lines of force. Let us draw the magnetic lines of force by using a compass by performing the following activity.

Activity -2

To draw the magnetic lines of force when the North Pole of the magnet is aligned with the geographic North.

Items required: Drawing board, white paper, pins, magnetic compass, bar magnet, iron filings etc.

Process:

1. Keep the drawing board on a table.
2. Fix a white paper on the board with pins, or cello-tape.
3. Remove any magnetic materials nearby.
4. Draw a line at the centre of the paper along its length and place a small compass on it. Now move the board till the compass needle aligns itself straight along the drawn line. Now, this line is aligned with the earth's North-South direction. Write N and S at the ends of the line.

5. Place a bar magnet straight on the line so that the line passes through the magnet centre, as shown in figure-3(a). Now the North Pole of the magnet is facing the geographic North. Draw an outline of the magnet.

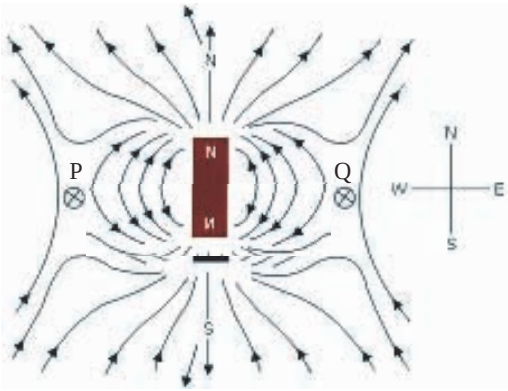


Figure-3(a): Drawing magnetic lines when the North

6. Draw a point near the magnet's North Pole, from where you shall start drawing the lines of force. Place the compass on this point in such a way that the South end of the compass needle lies in a straight line with this point. Now draw another point with the pencil at the other (North) end of the compass needle. Repeat the above by again placing the south end of the compass needle at the second point and draw another point at the other needle end. In this way, move the compress forward to draw the various points, till you reach the South Pole of the magnet.
7. Now draw a thin line through all the points mapped from the magnet's North Pole to the South Pole.
8. Mark direction arrows along the line pointing towards the South Pole. This is a magnetic line of force.
9. Similarly, draw other lines of force, starting from various points near the magnet's North Pole and ending at the South Pole. These lines of force will look like those shown in figure-3(a).

We observe that the lines of force from a curved square shape around points P and Q. When we place the magnetic compass in this square and bring an iron nail near it, the compass needle is attracted towards the nail. On removing the nail, if the compass needle stays stationary in the same position, this, indicates the absence of any magnetic field. This is known as a neutral point of the field.

Neutral point

We know that the earth acts like a magnet and therefore, it also has a magnetic flux area. When a magnet is placed anywhere, the area around it has two magnetic fields. First is the magnet's own magnetic field and second is the earth's magnetic field. The magnetic lines of force in this area are a resultant of the two fields. Any point in this area, where the magnet's line of force a equal and opposite to the earth's magnetic lines of force will be a neutral point. Point P and Q in figure 3(a), are two such point.

The magnetic force lines of various states of a magnet are shown in figure-3(b), 4(a), 4(b), 4(c) and 4 (d).

Magnetic lines of force are a vector quantity and their direction is taken the same as that a unipolar North Pole will follow in the magnetic field. Therefore, the magnetic lines of force appear at the North Pole and enter into the south. Inside the magnet, they go from south towards north. Hence, it forms a closed loop.

These lines of force do not cross or intersect each other, because if they did it would result in the campus needle pointing in two directions simultaneously at the intersection point, which is not possible.

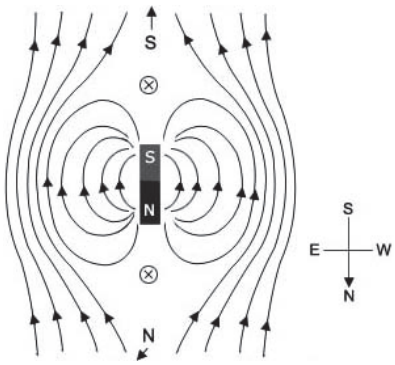
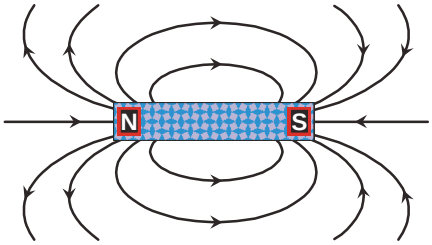
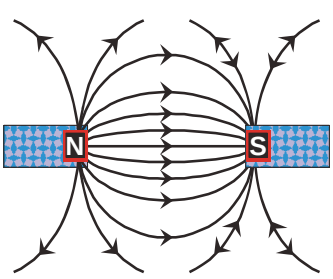


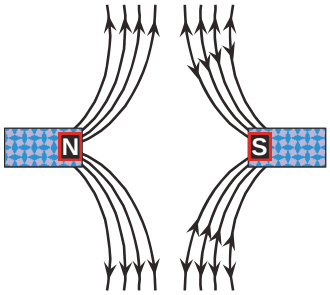
Figure-3(b): Magnetic lines of force when magnetic North Pole is aligned to geographical South



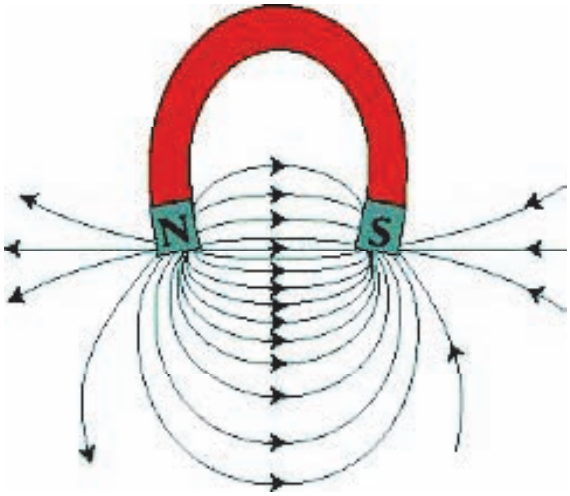
(a) Magnetic lines force of a bar magnet



(b) field lines when opposite poles of two magnets face each other



(c) field lines when similar poles of two magnets face each other



(d) field lines of horseshoe type magnet

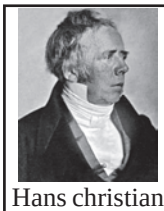
Figure-4 (a), (b), (c), (d): Magnetic lines of force

Questions

1. What are magnetic lines of force?
2. Write the properties of magnetic lines of force

12.2 Magnetic Field Due To Current Carrying Conductor

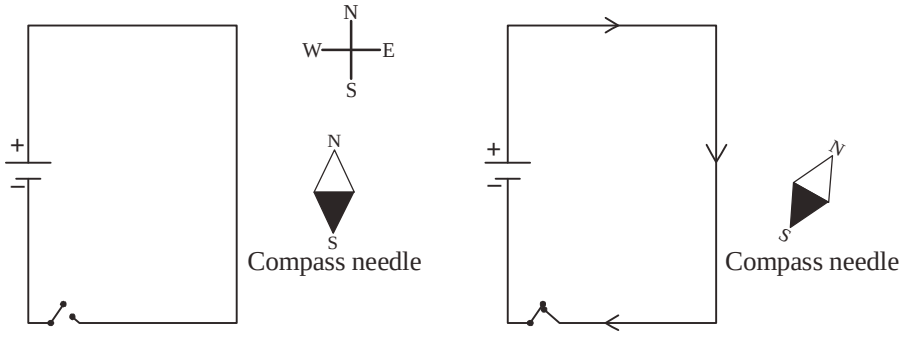
When electric current flows in a conducting wire, a magnetic field is produced around it. To observe this phenomenon, take a conducting wire (copper, aluminium etc.). Join the two ends of the wire to the terminals of a battery. Place a compass near and parallel to this wire (figure-5a). On switching on the circuit, current starts flowing in the wire and the compass needle gets deflected. This shows that a magnetic field is produced when current flows in a



Hans christian orsted

Electro-magnetic effects were first observed by Danish scientist H.C. Orsted in 1810.

conductor. The needle deflection intensifies with increase in the current and the direction of the magnetic lines of force gets reversed if the direction of current flow in the circuit is changed. If the current is switched off, the magnetic field also disappears (becomes null). Therefore, the magnetic field around the conductor is the result of the flow of current.



(a) No deflection when current is off (b) Needle deflects when current flows

Fig 5: Deflection of the compass needle due to flow of current

12.3 Magnetic Field Produced in a linear current carrying Wire

When a current carrying wire produces a magnetic field around itself, how is the shape of the magnetic field determined? Does the shape of the magnetic field depend upon the shape of the conductor? To test this, let us do the following activity.

Activity-3

Material required one 6V battery, one way key/switch, rectangular cardboard piece, 1 piece of straight thick copper wire, iron filings, etc.

Method:

1. Take the piece of cardboard.
2. Pierce the copper wire, vertically through the centre of the cardboard, so that the cardboard remains fixed in a flat horizontal position.
3. Connect the wire to a battery and key as shown in figure-6.
4. Sprinkle iron filings around the wire and place a small compass on the edge of the cardboard.
5. Now, switch on the current by closing the circuit, and gently tap the cardboard. You will observe that the iron filings arrange themselves in a particular shape and the compass needle gets deflected in a particular direction.

The iron filings arrange themselves around the wire in concentric circles. You can draw the lines of force and mark the direction of the field with arrows. Now, repeat this activity by changing the direction of current flow in the wire. What do you observe?

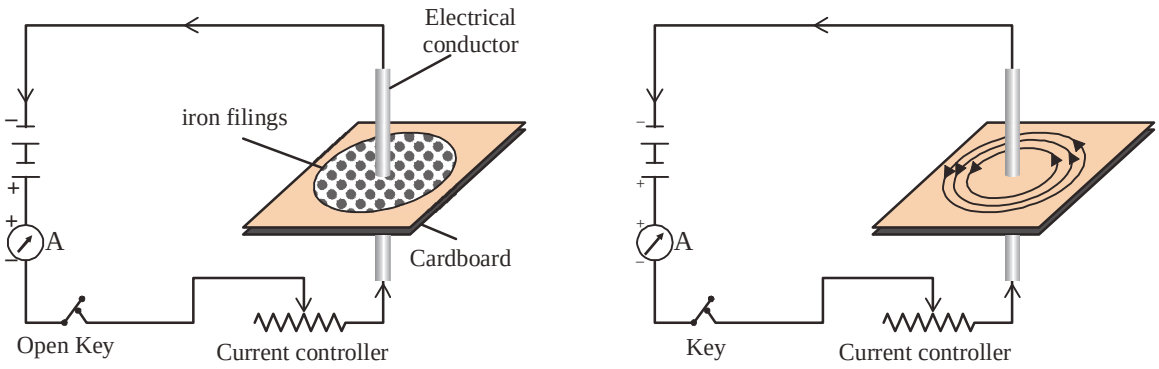


Figure-6: Magnetic field of a straight wire conductor

Now add a current controller and ammeter to the circuit as shown in figure-6. Change the amount of current flowing in the circuit and observe the effects on the magnetic field by placing the compass at various points on the cardboard. You shall observe that the needle deflection increases in intensity when the current flowing in the conductor increases. If the current flow is kept constant, but the compass is brought nearer to the conductor, the deflection in the compass needle becomes more pronounced, indicating higher field intensity. You will observe that the distance between the concentric lines of force is shorter near the conductor and the circles grow further apart and more dispersed as the distance from the conducting wire increases. Therefore, the magnetic field intensity is inversely proportional to the distance from the current carrying conductor.

12.4 Right Hand Rule For Diretion Of Magnetic Field

The direction of the magnetic lines of force of a magnetic field produced by a straight current carrying conductor can be determined easily, by the right hand rule. Imagine that you place your right hand around the conductor in such a way that the thumb points towards the direction of current flow, then the fingers bent around the conductor will point in the direction of the lines of force, as shown in figure-7.



Figure-7: Right hand rule

Questions

1. Draw the magnetic lines of force produced by a linear conductor, carrying electric current.
2. If in a conducting wire, current is flowing from east to west, what would be the direction of the lines of force at a point, vertically below the wire and at a point above that wire?

12.5 Magnetic Field Due To Circular Current Carrying Conductor

We have seen the shape of the magnetic field produced by a straight current carrying conductor. Imagine that the straight wire is now bent into a circular coil, and current flows through it. What would be the shape of the magnetic field produced?

Let us fix a circular coil on a cardboard sheet as shown in figure-8. When current starts flowing in the coil, sprinkle the iron filings on the cardboard. We can observe that the iron filings get arranged in concentric circles around the two strands of the loop figure-8. As we move away from the wire, the circles get larger in diameter and spaced further more apart.

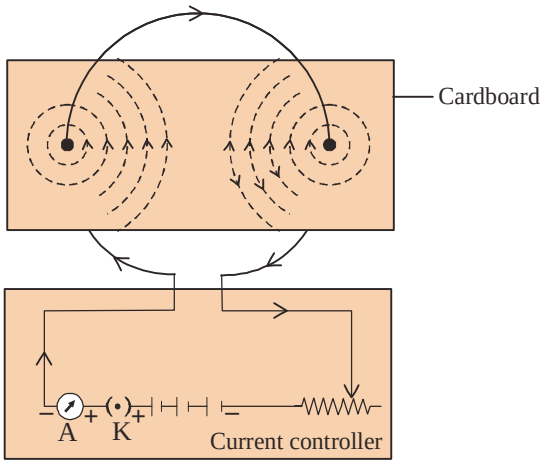


Figure-8: Magnetic field due to circular current carrying conduct

As the distance increases, the lines appear almost straight due to the large diameter.

By using the right hand rule, we can observe that the circular wire produces a magnetic field which has the lines of force in the same direction at every section of the circular coil, and therefore, the intensity of the magnetic field gets magnified. Let us perform an activity to observe this phenomenon.

Activity-4

Materials required- circular cardboard piece, insulated copper wire coiled into about 50 loops, current controller, battery (6V), switch, ammeter, etc.

Method:

1. Take the circular cardboard and make two holes in it.
2. Now pierce the copper wire coil through the holes in such a manner that the cardboard lies horizontally on the vertically held loop.
3. Join the battery, current controller, ammeter, and switch in series in this circuit as shown in figure-8.
4. Sprinkle iron fillings uniformly on the cardboard
5. Switch on the current.
6. Gently tap the cardboard to enable the iron filings to arrange themselves in a particular shape and study the pattern formed.

We have learnt that the intensity of the magnetic field produced by a current carrying conductor is directly proportional to the amount of current flowing in the conductor. If we have a wire coil of n circular loops, then the magnetic field intensity would be n times that produced by a single coil. This is due to the fact that the direction of the current flowing in each loop is same.

Now repeat the above activity by reducing the diameter of the wire loop. The coil acts as the thin disc magnet, where one plane of the disc acts as a North Pole and the other plane as the South Pole of the magnet.

12.6 Magnetic Field Due To A Solenoid

A solenoid is a cylindrical shaped tube formed by tight coils of conducting copper wires. Figure-9 shows a solenoid whose two ends are joined to a battery. When current flows in the solenoid, a magnetic field is produced around the solenoid. Compare the pattern of magnetic field in figure-9 to the magnetic field shape shown in figure-4 (a).

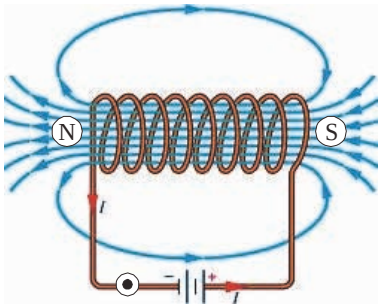


Figure-9: Magnetic field of a current carrying solenoid

In both the figures, the magnetic line of force appear as similar in shape and pattern and thus, the magnetic field produced by a solenoid is similar to that of the bar magnet, and one end of it acts as a North Pole while the other end acts like a South Pole. The magnetic lines of force inside the solenoid are parallel lines which show that the magnetic field inside the solenoid is uniform. The magnetic field produced inside the solenoid can be used to produce a magnet by placing a piece of magnetic material like iron, inside the solenoid, as in figure-10 (a).

This type of magnet produced is known as electromagnet.

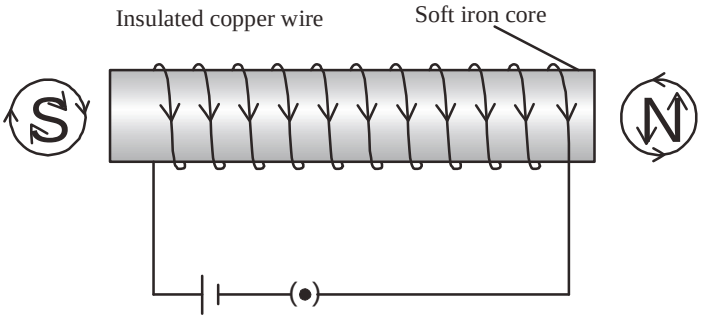


Figure-10 (a): electromagnet

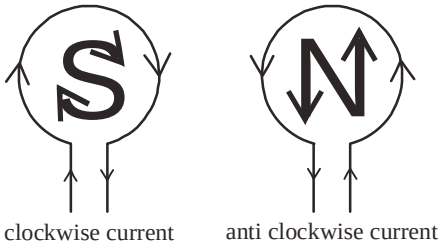


Figure-10 (b): electromagnet

The circular shape formed by bending a conducting wire is called a loop or coil. The coil end where the current direction is clockwise acts as a South Pole, and the other end where the current direction is anticlockwise acts as the North Pole.

Questions

1. Imagine a circular loop of wire placed on a table. If a clockwise current is flowing in the loop, find the direction of the magnetic lines of force inside the loop and outside the loop.
2. Draw an example of magnetic field where the lines of force are uniform.

12.7 Force On Current Carrying Conductor In A Magnetic Field

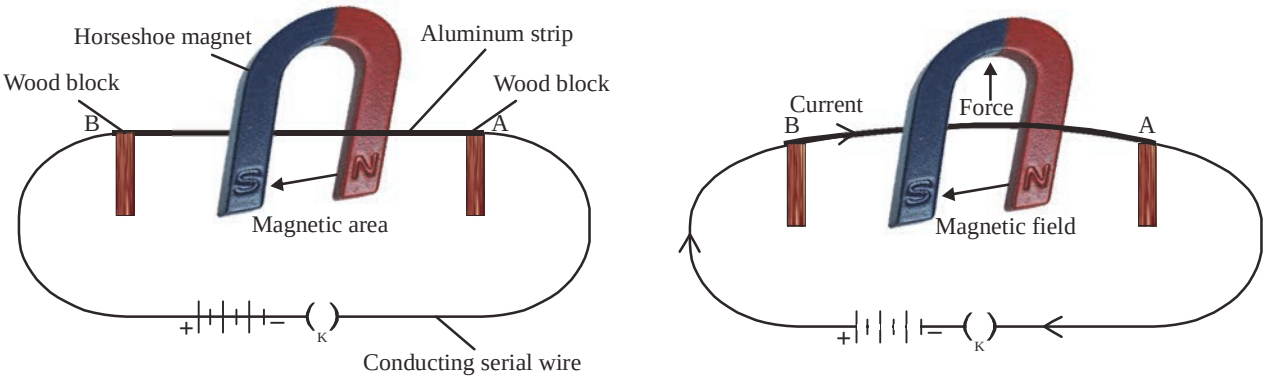
We learnt that when electric current flows in any conductor, it produces a magnetic field around the conductor, and when a compass is placed in this field, the compass needle experiences a force. Similarly, the compass will produce an equal but opposite force on the conductor.

To study the force affecting a current carrying conductor due to presence of a magnetic field, let us conduct the following activity-

Activity-5

Materials required: small aluminium rod AB, joining wires, horse shoe magnet, battery, switch, wooden stand.

Method: Fix the aluminium rod on the wooden stand with the help of joint wire as shown in figure-11. Place a strong horse shoe magnet in such a way that the aluminium rod lies centrally and horizontally between the two poles of the magnet. The North Pole of the magnet should be at the front of the rod and South Pole in the back. Due to this, the direction of the magnetic field lines is from left to right of the rod.



(a) (b)

Figure-11 (a), (b): Force on current carrying wire

- Now switch on the current in the aluminium rod (B to A)
- What do you observe when the current flows?
- We observe that the rod turns left.
- Now change the direction of current flow and write your observations.

The motion of the conductor in this activity shows that a force is exerted on the current carrying conductor when placed in a magnetic field. The direction of this force changes on changing of the current direction.

Now change the position of the magnetic poles in the above activity and observe whether the direction of the force also changes. We observe that the force direction changes on changing the direction of the magnetic field. This shows that the direction of the force exerted on the conductor depends both on the direction of the magnetic lines of force as well as the direction of the current flowing in it.

In this activity, we kept the current direction and magnetic lines of force perpendicular to each other and observed that the direction of force produced is also perpendicular to both of them. To indicate this, Fleming gave the left hand rule.

12.8 Fleming's Left Hand Rule

When a current carrying conductor is placed in a magnetic field the conductor experiences a force. Fleming gave the left hand rule, to determine the direction of the force experienced by a conductor when placed in a magnetic field. A left hand can be held as shown in figure 12, where the thumb, the first finger and middle finger are held perpendicular to each other. The thumb represents the direction of motion resulting from the force, experienced by the conductor II the first finger represents the direction of the magnetic field, then the middle finger represents the direction of current.

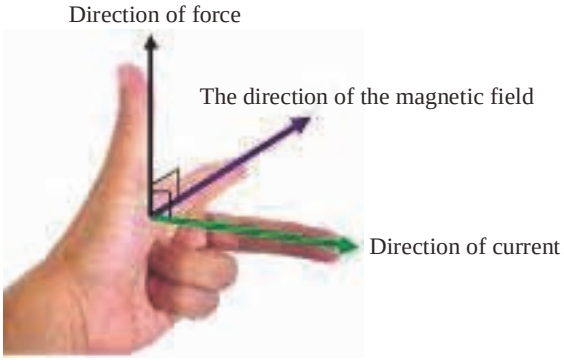


Figure-12: Fleming's Left Hand Rule

Electric motors and electric generators are some of equipments, which make use of magnetic field of a current carrying conductor.

Questions

1. In activity-5, how will the motion of aluminium rod AB be affected when,
   (a) The magnitude current in the rod is increased.
   (b) A more powerful horse shoe magnet is used.
   (c) The length of the rod is reduced.

12.9 Electric Motor

An electric motor is a device which converts electrical energy into mechanical energy. They are used in numerous appliances and machines like electric fans, refrigerators, washing machines, water pumps etc. Electric motors can be powered by direct current (D.C.) power sources such as batteries, rectifiers etc. or by alternating current (A.C.) sources such as power supply grids, generators, etc.

Principle: when a circular coil is placed inside a magnetic field and current flows in the coil, then force acts on the coil, which moves (rotates) it continuously. When the coil in turn, rotates the axle (shaft) attached to it, and the axle delivers the mechanical power. In this manner, a motor converts electrical energy into mechanical energy.

Construction of motor and it working principle

In an electrical motor, as shown in figure-13, a rectangular loop (coil) of insulated copper wire ABCD, is placed between two cylindrical poles of a permanent magnet in such a way that the arms AB and CD are perpendicular to the magnetic field lines of force. The loop can be made to spin by attaching a half circle of copper, X and Y, to each end of the loop. The inside surface of the copper half circles is insulated, and fixed on the shaft (axle) of the motor and the shaft rotates along with the

loop. The outer surface of the copper half circles are joined to two stationary (fixed) conducting carbon brushes P and Q, as shown in figure-13.

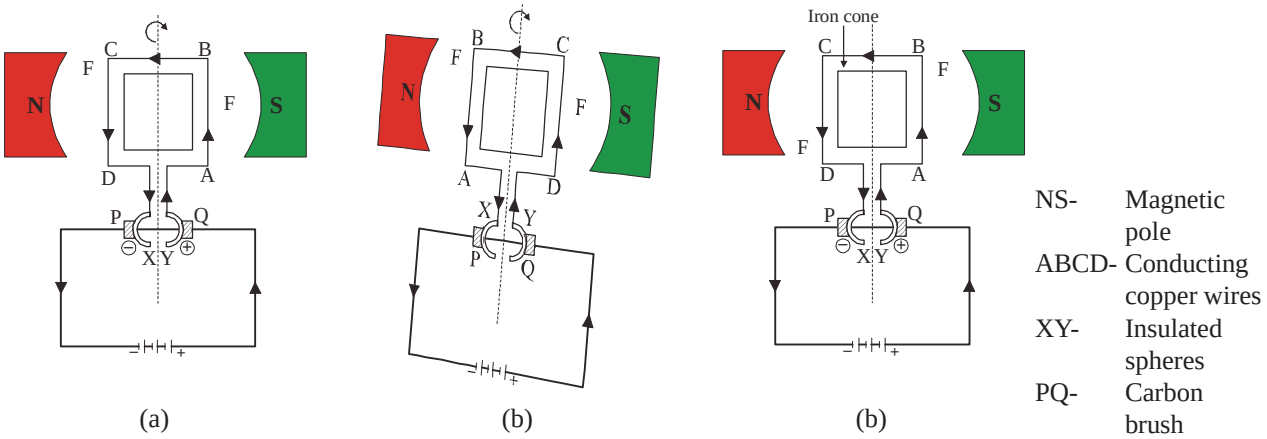


Figure-13: Simple electric motor

Working: The current flowing from the battery enters the loop through the brush Q and flows out through the brush P to complete the circuit. Note that in loop ABCD current flows from A to B, B to C and C to D. Therefore, the current flows in opposite directions in the two arms, AB and CD. Due to this the resulting force generated, acts on AB in a downward direction and CD in an upward direction. Therefore, the loop turns or rotates in an anti-clockwise direction. After half rotation, the half circle Y comes in contact with fixed brush P, and half circle X is in contact with brush Q. thus the direction of current flowing in the loop remains the same, but gets reversed in the loop arms. The current now flows from DC to BA. This arrangement to reverse to reverse the flow of current in loop arms is known as a dual changer. The two half circles act as a dual changer. Due to this reversal, the loop continues to rotate in an anti-clockwise direction. In heavy motors, used in industries, the permanent magnets are replaced by electromagnets. Increasing the number of turns in the coil, increases the amount of current which in turn increases the power of the motor. The coil is wound tightly around a soft iron piece known as core. The coil and core assembly is known as armature.

12.10 Eletromagnetic Induction

In this chapter, we have studied that when a current carrying conductor is placed in a magnetic field, it experiences a force. This force results in the motion of the conductor. Now, imagine a situation, where if a current carrying conductor is moved through a magnetic field, an electric current is produced in the conductor. This phenomenon was first studied by Michael Faraday. Let us perform an activity to observe this.



Faraday is known for Electro-magnetic induction, force fields and laws of electro-magnetism

Michael Faraday

Activity-6

Materials required: a copper coil of numerous turns, galvanometer, a strong bar magnet.

Method: 1. Take a coil PQ, having numerous turns of copper wire.

2. Join the two ends of the coil to a galvanometer as shown in figure 14.

3. Take a strong bar magnet and move its north pole towards the coil end Q. do you notice ant deflection in the galvanometer needle.

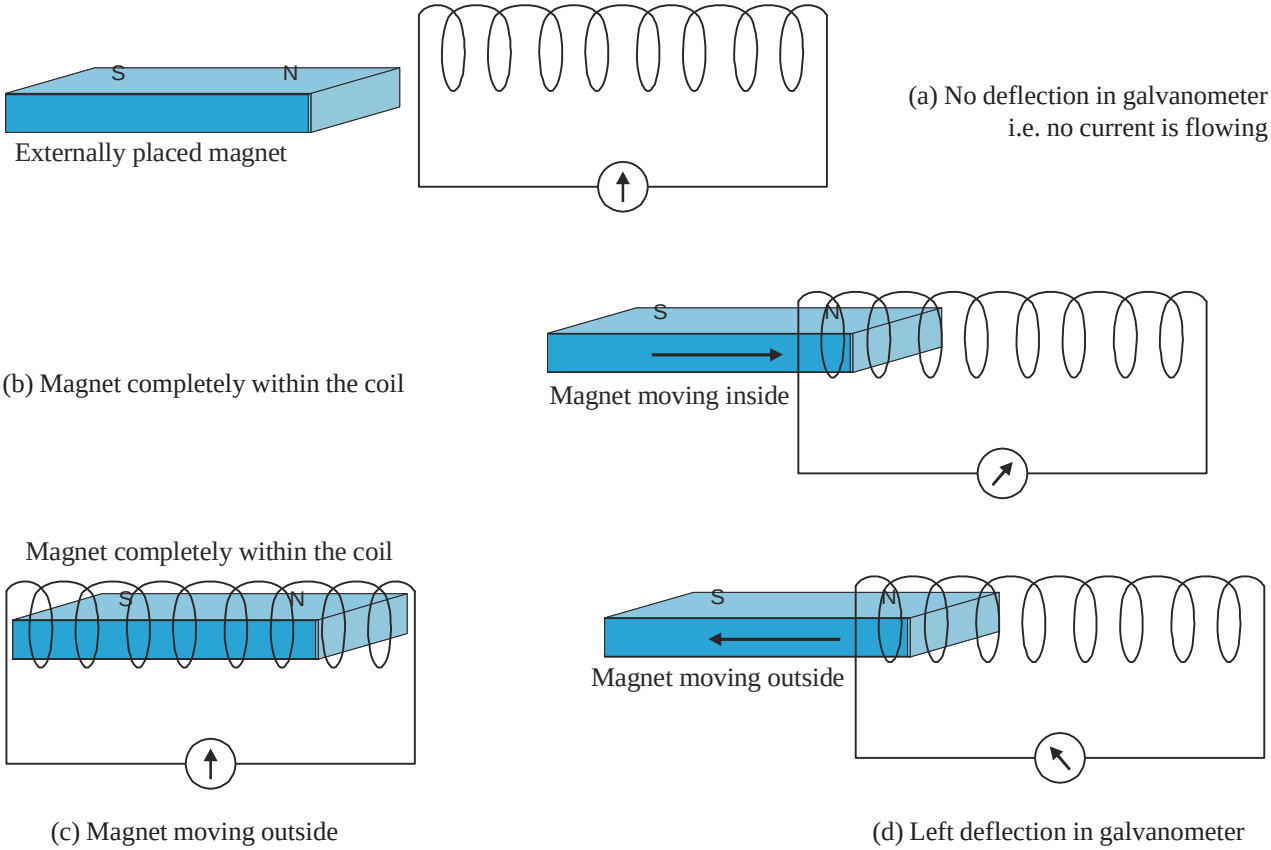


Figure-14: Indication of electro-magnetic

4. You will notice an instant deflection in the needle of galvanometer which indicates the flow of current in coil PQ.

5. As soon as the movement of the magnet is stopped, the deflection of the needle is nil.

6. Now, move the North Pole of the magnet away from the coil Q and notice that current again flows in the coil but now the deflection of galvanometer needle is in opposite direction.

You can also observe that if the south pole of the magnet is moved towards coil end Q, the deflection of the galvanometer is in opposite direction to that when the north pole of the magnet was moved towards Q. This activity, demonstrates that when a conductor is moved in a magnetic field, or the magnet is moved, current is produced in the conductor and this is known as an induced current. This phenomenon is known as electro-magnetic induction.

Flemming's right hand rule is used to find the directijon of current in this activity.

12.11 Fleming's Right Hand Rule

According to this rule, spread the thumb, the first finger and the middle finger of the right hand in such a way that they are perpendicular to each other., as shown in figure 15.

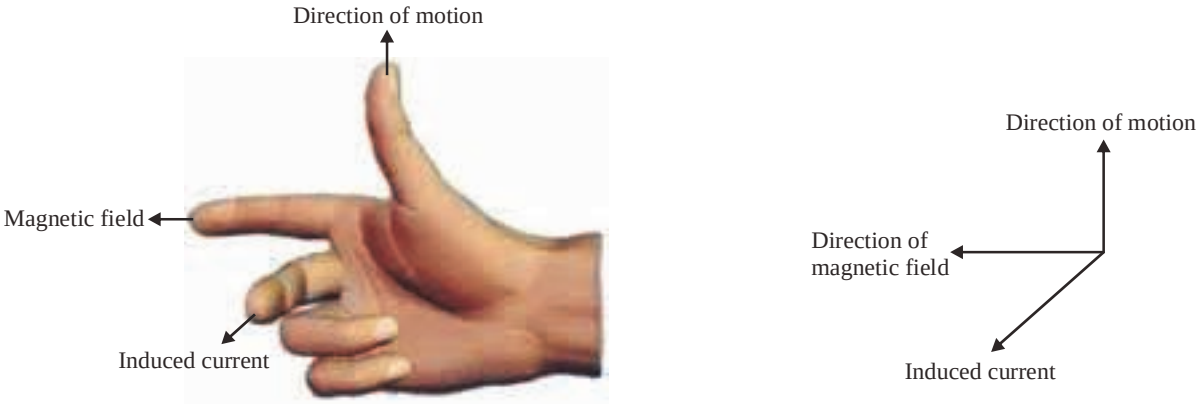


Figure-15: Fleming's right hand rule

The thumb represents the direction of motion of the conductor, first finger shows the direction of magnetic field lines of force and the middle finger points in the direction of the flow of induced current.

12.12 Dynamo Or Eletric Generator

Dynamo is an apparatus, which converts mechanical energy into electrical energy. It works on the principal of electromagnetic induction. This phenomenon is also used to generate large scale electric power by power plants for use in home and industries.

Construction:

1. Armature: for this, a rectangular coil having numerous turns of insulated copper wire is tightly wound around a soft iron core, as shown in figure-16.
2. Magnet: the armature is placed between the insides of the NS poles of a strong horse shoe magnet.

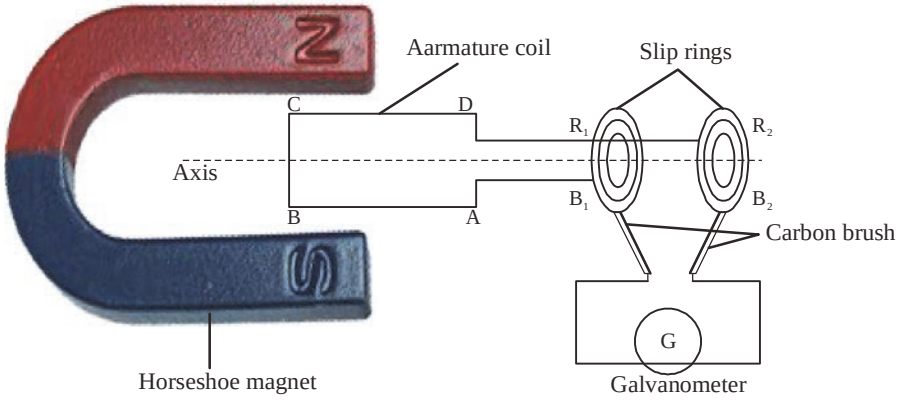


Figure-16: Electric generator

3. Circular dual changer: R_1 and R_2 are metallic half circles which are mounted on a shaft.
4. Carbon brush: fixed current conducting carbon brushes B_1 and B_2 are placed touching the half circles R_1 and R_2 respectively. The brushes are joined to a galvanometer.

Process: when the armature shaft attached to two half circles is moved in such a way that arm AB above the coil is above the arm CD, then the intensity of the magnetic lines of force passing through the loops varies and an induced current is produced in the two arms AB and CD, of the loop, as given by Fleming's left hand rule. The current generated in each coil of the armature would add up, and create a powerful current in the loop. In the external circuit, the current will flow from the brush B_1 towards B_2. After half rotation of the loop, the arm CD is now above the arm AB and the induced current will flow in the direction DCBA. The half circle R_2 is now in contact with brush B1 and therefore the current flows from B_1 towards B_2. The current in this arrangement flows in the same direction during rotation of the loop and it is known as direct current generator.

To obtain alternating current (A.C.), we use instead of half circles, a circular ring (slip ring), due to which the AB arm of the armature is always joined to brush B1 and CD arm to the brush B_2. Therefore, when the direction of the current in the loop is ABCD, then in the external circuit it flows from B_2 towards B_1 and when the direction is DCBA, then in the external circuit, it flows from B_1 towards B_2.Thus, the direction of current reverses with every half rotation of the loop. This type of current, which periodically reverses its direction, is known as alternating current or A.C. The assembly to produce such type of electricity is known as alternating current generator.

Questions

1. What is an electric generator?
2. Write the principle on which an electric generator works.
3. What is a circular dual changer?

Keywords

Dual pole magnetic compass, magnetic field, straight conductor, magnetic lines of force, solenoid, neutral point, dynamo or generator, alternating current, direct current, half circle dual changer, slip ring, Maxwell's rule, Fleming's left hand rule,Fleming's right hand rule.



What We have learnt

- Compass needle is a small magnet whose one end always points towards North direction and is known as North Pole. The other end pointing towards south is called the South Pole.
- The magnetic axis and the geographical axis of earth are bent at an angle of 17° to each other.
- The intensity of magnetic flux surrounding a magnet indicates the power of the magnet.

- The magnetic field is represented by magnetic lines of force. A line of force is a circular curve along the tangent of which an imaginary unipolar North Pole travels.
- The lines of force are concentrated where the intensity of magnetic field is high.
- Two lines of magnetic force never intersect each other.
- At the neutral point, the magnetic field of the magnet is equal but opposite to the earth's magnetic field.
- A magnetic field is associated with a current carrying conductor. The lines of force form concentric circles around the conductor and their direction is determined by the left hand rule.
- Solenoid is a coil formed by numerous turns of insulated copper wire.
- The electromagnet has a core of soft iron around which a coil of insulated wire is wound tightly.
- When a current carrying conductor is placed in a magnetic field, it experiences a force, and the direction of this force is determined by Fleming's right hand rule.
- Electric motor is a device to convert electrical energy into mechanical energy.
- Electric motor works on the principle of force created on a conductor by a magnetic field.
- Electromagnetic induction is a phenomenon where induced generated by varying the intensity of magnetic field in a conducting coil. The direction of induced current is determined by the Right hand rule of Fleming.
- Electric generator or dynamo converts mechanical energy to electrical energy.
- Dynamo works on the principle of electromagnetic induction.

Exercise

1. Choose the correct option

   (i) The magnetic lines of force determine,

   (a) shape of the magnetic field (b) direction of the field

   (c) intensity of the magnetic field (d) intensity and direction of magnetic field

   (ii) A straight current carrying conductor has a magnetic field

   (a) lines of force parallel to the conductor

   (b) lines of force perpendicular to the conductor

   (c) lines of force in concentric circles around the conductor

   (d) lines of force starting radially from the wire end

   (iii) A solenoid has a magnetic field inside it, which is

   (a) different at each point (b) same at each point

   (c) zero (d) None of the above

(iv) electromagnetic induction is a phenomenon to

(a) charge any matter (b) rotate a coil
(c) produce magnetic field in a coil
(d) to produce induced current when either coil or magnet moves.

(v) We can generate electric current using–

(a) Generator (b) Motor
(c) Galvanometer (d) Ammeter

2. Fill in the blanks

(i) That device which converts mechanical energy into electrical energy is .....

(ii) ..... converts electrical energy into mechanical energy.

(iii) Fleming's left hand rule indicates the direction of .....

(iv) Fleming's right hand rule indicates the direction of ..... acting on the conductor.

(v) Maxwell's left hand rule indicates the direction of .....

3. Which scientist gave the principle of electromagnetic induction?

4. A magnetic field is created around a conductor when current flows in it. Which scientist first confirmed this phenomenon?

5. List three appliances that use electric motor.

6. Write the right hand rule of Fleming.

7. List the three ways in which a magnetic field can be produced.

8. How does a solenoid act as a magnet? Can you determine its North and South poles by help of a bar magnet?

9. Write the three chief properties of a magnet.

10. What are magnetic lines of force? Write the three main properties of lines of force.

11. Why two magnetic lines of force do not intersect each other?

12. Determine the electric motor under the following headings,

(a) labelled diagram (b) working principle (c) working process

13. Describe the electric generator under the following headings,

(a) labelled diagram (b) working principle (c) working process

14. Two circular coils A and B are kept near each other. If the current flowing in coil A is varied, would a current be induced in coil B. Answer giving reasons.

15. Connect an insulated copper coil to a galvanometer. What will happen if a bar magnet is,

(a) moved inside the coil (b) kept stationary in the coil (c) moved outwards from inside the coil

16. Write the rule which represents the direction of magnetic lines of force produced by a straight conductor.

17. Make an electromagnet